Discussion Exercise 5

University of California, Riverside
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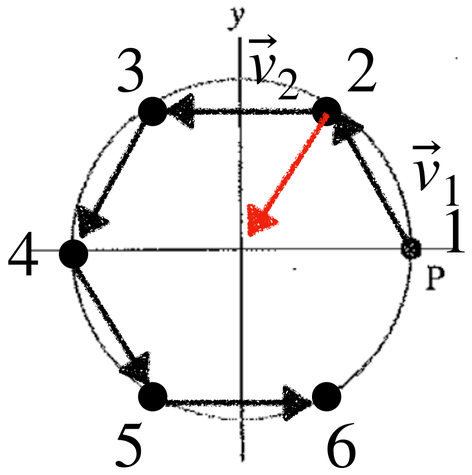
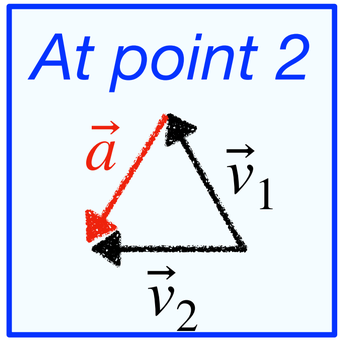
1. Position, Speed and Velocity in Circular Motion

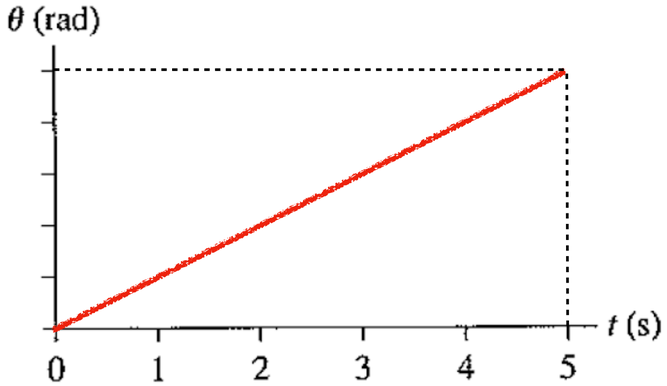
A particle undergoes uniform circular motion with constant angular velocity $\omega = +1.0$ rad/s, starting from point P.

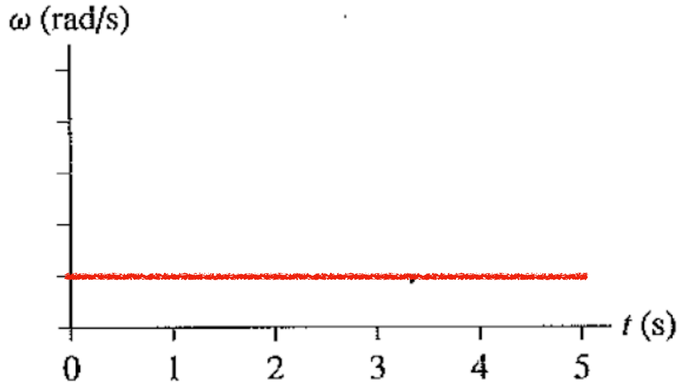
a. On the figure, draw a motion diagram showing the location of the particle every 1.0 s until the particle has moved through an angle of 5 rad. Draw velocity vectors **black** and acceleration vectors **red**. For this question, you can use 1 rad $\approx 60°$.

b. Below, graph the particle's angular position θ and angular velocity ω for the first 5 s of motion. Include an appropriate vertical scale on both graphs.







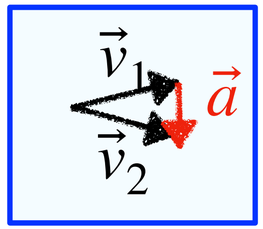


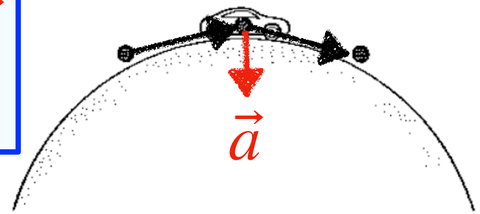
2. Apparent Forces in Circular Motion

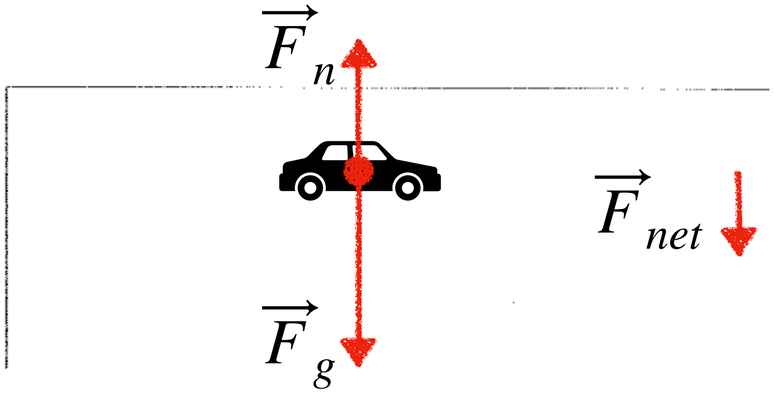
The drawing is a partial motion diagram for a car rolling at constant speed over the top of a circular hill.

a. Complete the motion diagram by adding the car's velocity vectors. Then use the velocity vectors to determine *and show* the car's acceleration $\vec{a}$ at the top of the hill.

b. To the right, draw a free-body diagram of the car at the top of the hill. Next to the free-body diagram, indicate the direction of the net force on the car or, if appropriate, write $\vec{F}_{net} = \vec{0}$.







$\vec{F}_{net}$ points downward.

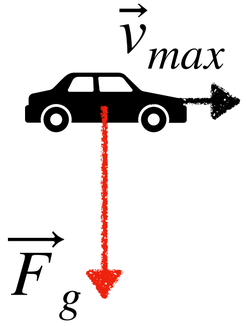
Notice that the direction of the acceleration in uniform circular motion always points to the center of the circular orbit.

c. Does the net force point of your free-body diagram point in the same direction you showed for the car's acceleration? *Yes* If not, you may want to reconsider your work thus far because Newton's second law requires $\vec{F}_{net}$ and $\vec{a}$ to point the same way.

d. Is there a maximum speed at which the car can travel over the top of the hill and not lose contact with the ground? If so, show how the free-body diagram would look at that speed. If not, why not?

Yes, there is.

By Newton's second law, $F_g - F_n = mv^2/r$. The maximum speed is reached when the normal force vanishes (why?), thus $F_g = mv_{max}^2/r \rightarrow v_{max} = \sqrt{F_g r/m} = \sqrt{gr}$. If the speed keeps increasing, then the car will leave the ground



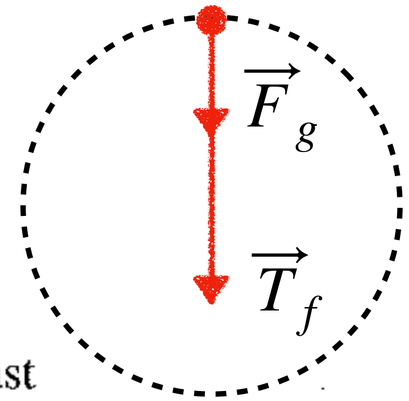
3. Forces in Circular Motion

You can swing a ball on a string in a *vertical* circle if you swing it fast enough.

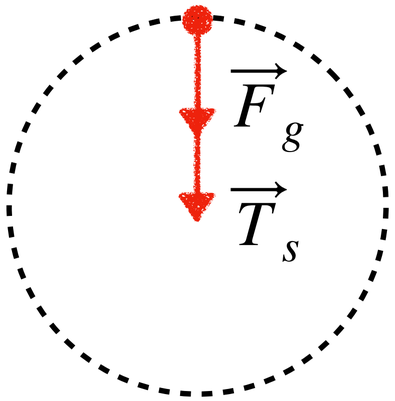
a. Draw two free-body diagrams of the ball at the top of the circle. On the left, show the ball when it is going around the circle very fast. On the right, show the ball as it goes around the circle more slowly.

Make sure that $T_f > T_s$.

Compared to the slower case, the faster case has a greater centripetal acceleration (recall that $a = v^2/r$), thus the tension force is greater in the fast case by Newton's second law ($F_g + T = mv^2/r$).



Very fast



Slower

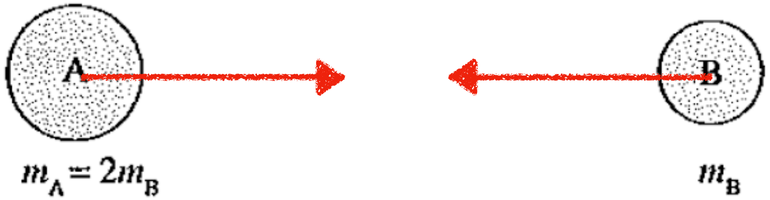
b. Suppose the ball has the smallest possible frequency that allows it to go all the way around the circle. What is the tension in the string when the ball is at the highest point? Explain.

The tension should be zero. Any finite tension pointing downward would increase the net force, thus increase the acceleration and the velocity. If the tension points upward, one can further reduce the smallest possible frequency/velocity, but the tension in the string can never point upward (since you can only pull a string)!

4.Newton's Law of Gravity

Star A is twice as massive as star B.

a. Draw gravitational force vectors on both stars. The length of each vector should be proportional to the size of the force. $F_{B\ on\ A} = F_{A\ on\ B}$



b. Is the acceleration of star A toward B larger than, smaller than, or equal to the acceleration of star B toward A? Explain.

By Newton's second law, we have $F_{B\ on\ A} = m_A a_A$ for A and similarly $F_{A\ on\ B} = m_B a_B$ for B

By Newton's third law $F_{B\ on\ A} = F_{A\ on\ B}$, thus we have $m_A a_A = m_B a_B$.

Notice that $m_A = 2m_B$, then $a_A = a_B/2$